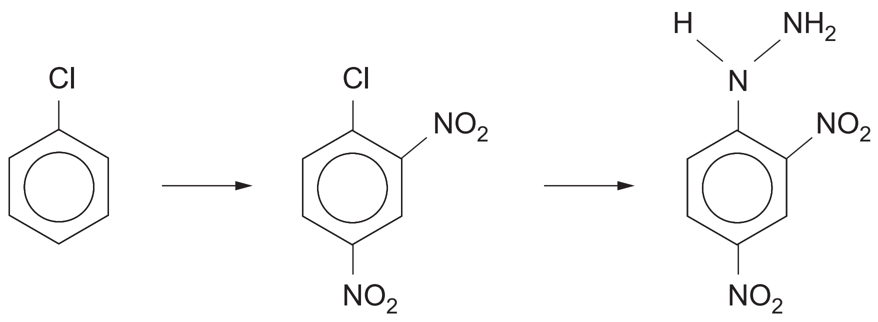
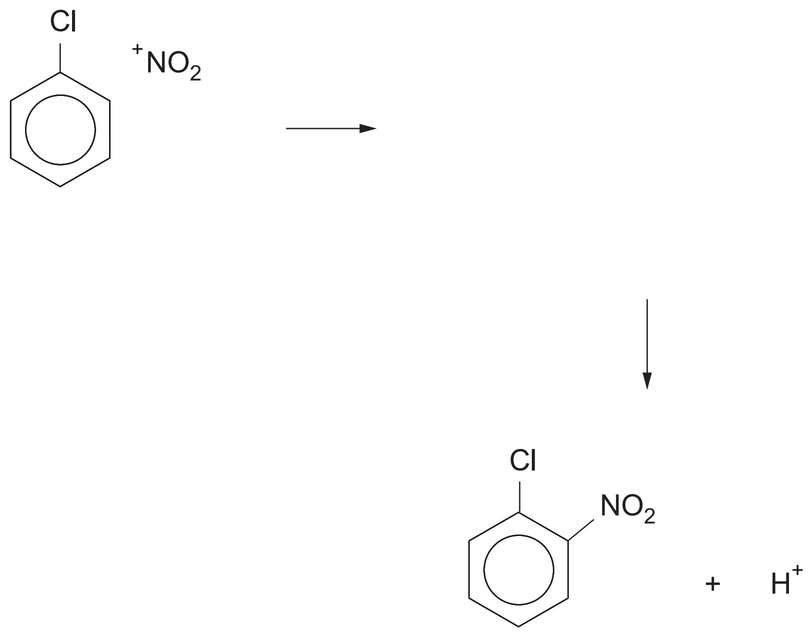
Examiner only

7. *(a)* 2,4-Dinitrophenylhydrazine (2,4-DNPH) can be made by the following route.



(i) Complete the mechanism for this reaction, which shows the nitration of chlorobenzene. You should use curly arrows and also show the structure of the intermediate. [2]



(ii) State the type of mechanism occurring in this nitration. [1]

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(b) State the feature of 2,4-DNPH that enables it to react as a base. [1]

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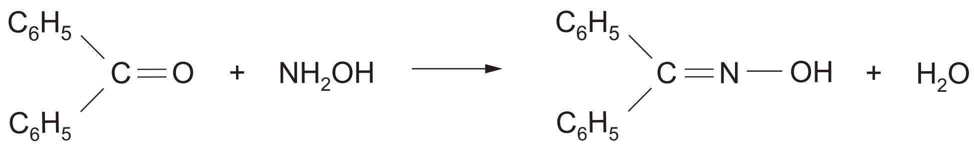
(c) 2,4-DNPH can be used as a reagent to identify aldehydes and ketones.

(i) State what is seen when 2,4-DNPH is used to identify an aldehyde or a ketone. [1]

(ii) Explain why 2,4-DNPH is an appropriate reagent to use in their identification. [1]

(d) Alternative reagents can be used for this identification. One of them is hydroxylamine, NH_2OH, which produces a white crystalline solid when it reacts with an aldehyde or ketone.

For example with diphenylmethanone



diphenylmethanone oxime
melting temperature 143 °C

In an experiment, impure diphenylmethanone oxime was recrystallised from flammable methanol (boiling temperature 65 °C).

Outline a **safe** procedure to obtain pure dry crystals of diphenylmethanone oxime using methanol as the solvent. You should assume that all the material present in the crude oxime is soluble in methanol. [5]

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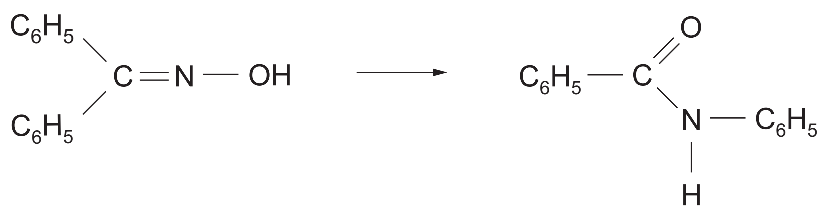
(e) Under suitable conditions diphenylmethanone oxime undergoes a rearrangement.



diphenylmethanone oxime N-phenylbenzamide

Describe how the infrared absorption spectra of these two compounds would differ, giving the relevant bonds and their wavenumbers. You should comment on **both** compounds when considering differences. [3]

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